

Theory of dynamical stability in attractor neural networks predicts CA3 neurons' activation functions

question

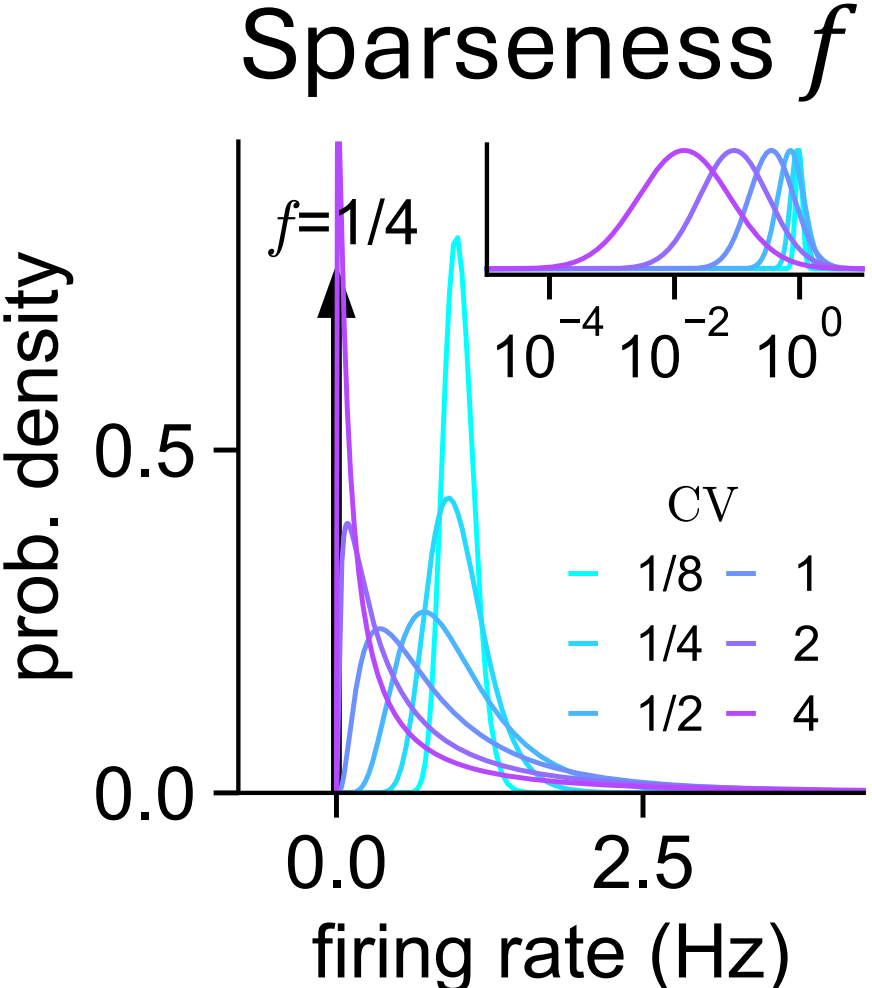
Under what conditions can you store rate-coded memory patterns as stable fixed points of a recurrent neural network?

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model

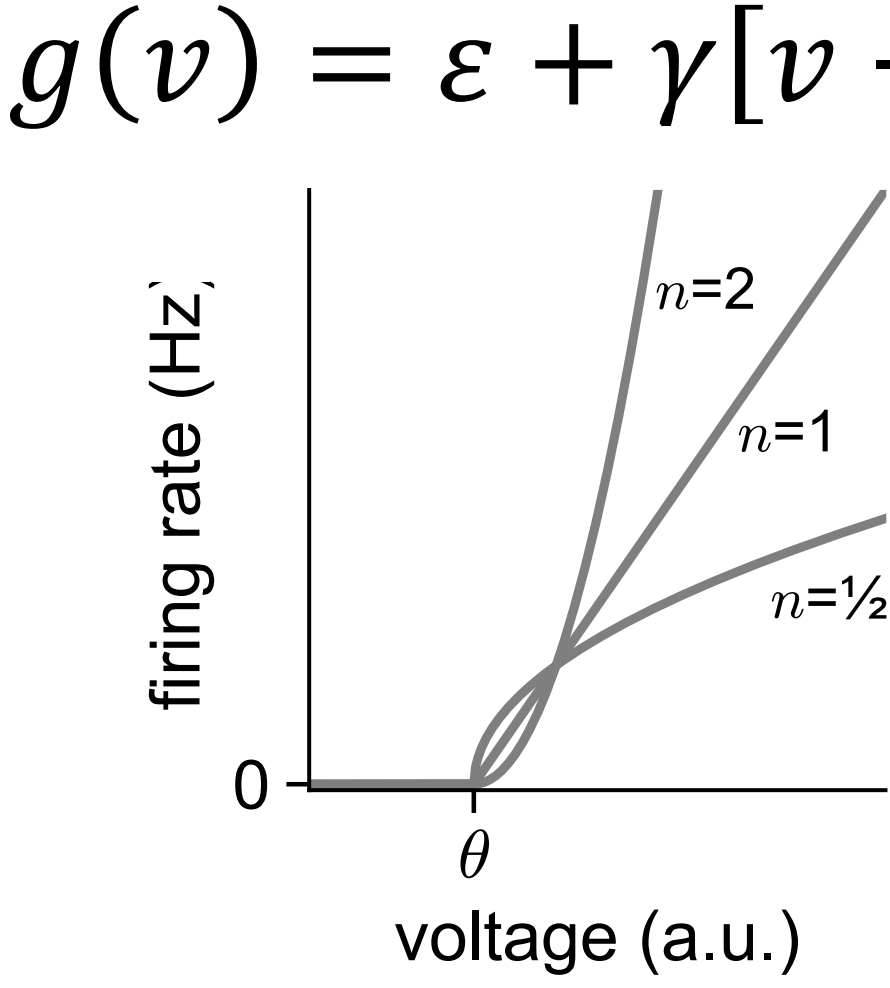
Sparseness f



Log-normal firing rate mean 1Hz, different CV

Graded patterns

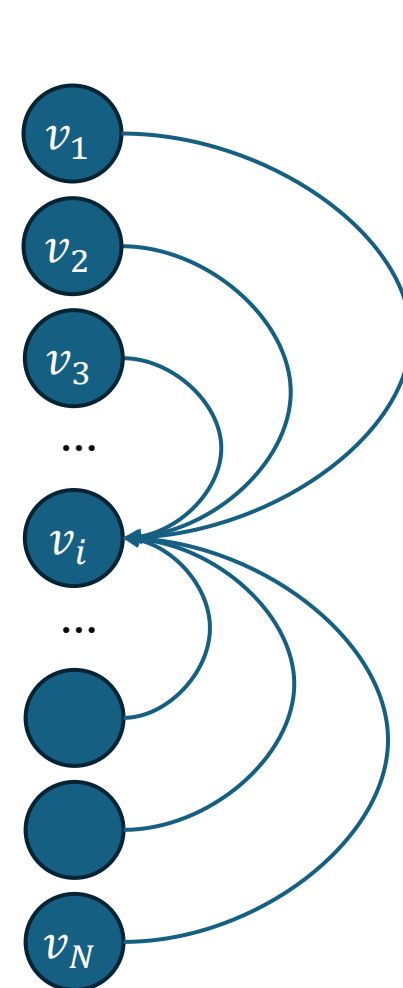
$g(v) = \varepsilon + \gamma[v - \theta]^n_+$



Neural activation

Voltage dynamics
 $\tau \dot{v} = -v + Wg(v)$

Rate dynamics
 $\tau \dot{r} = -r + g(Wr)$



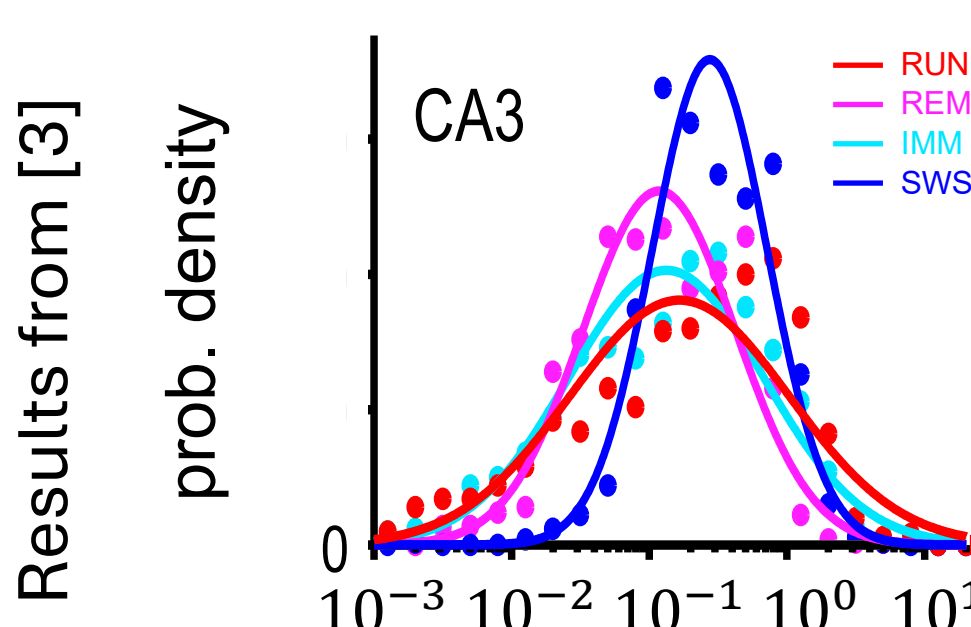
Recurrent networks

$W^* = \arg \min_W |W|$
s.t. $r^\mu = g(Wr^\mu)$
 $\forall \mu = 1 \dots P$

Storing patterns as network fixed points

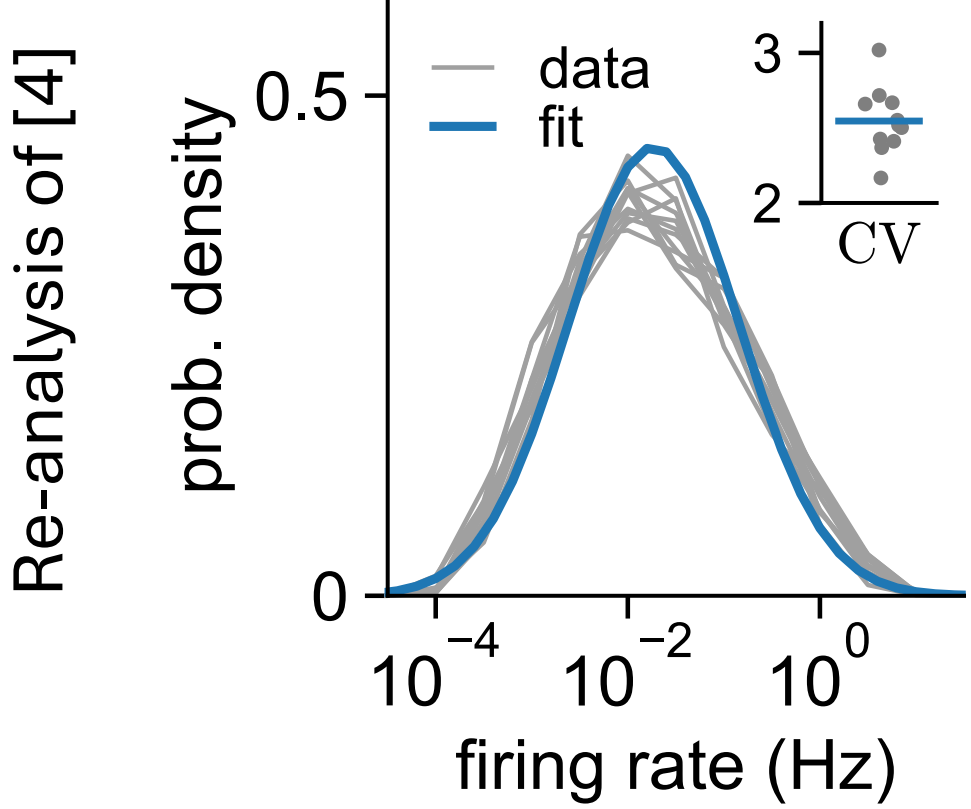
theory

Results from [3]



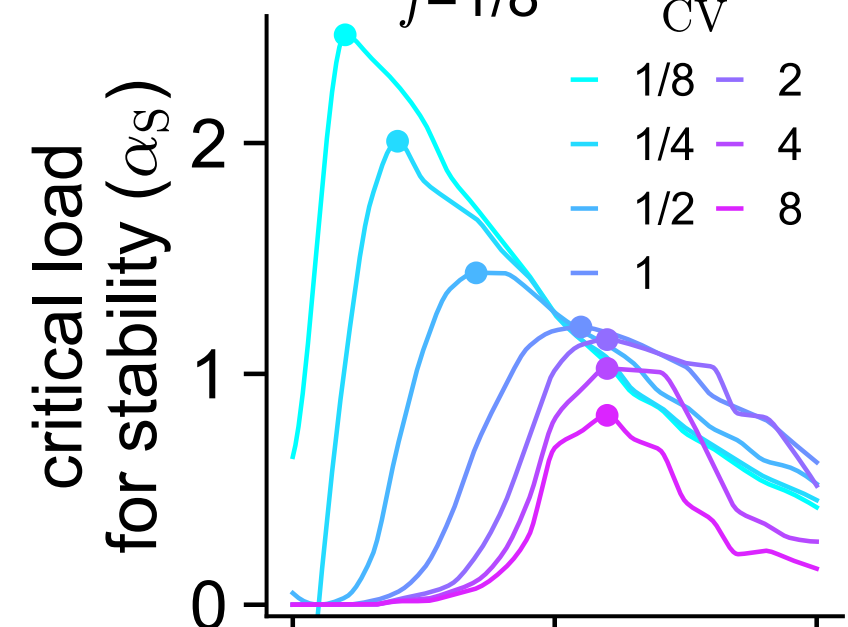
CA3 activity CV is 2-3 high-variance patterns

Re-analysis of [4]



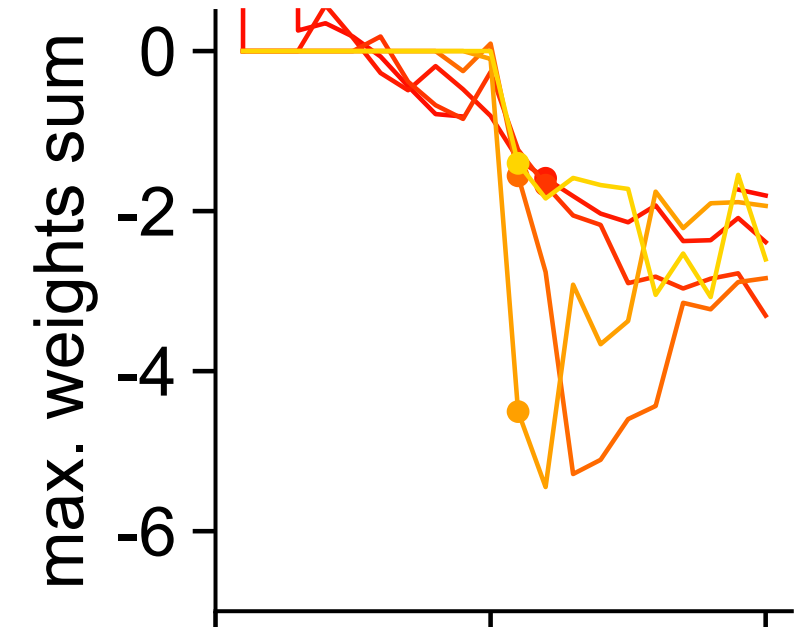
For CV ≥ 1, n^* is 1-1.2 threshold-linear activation

critical load for stability (α_S)



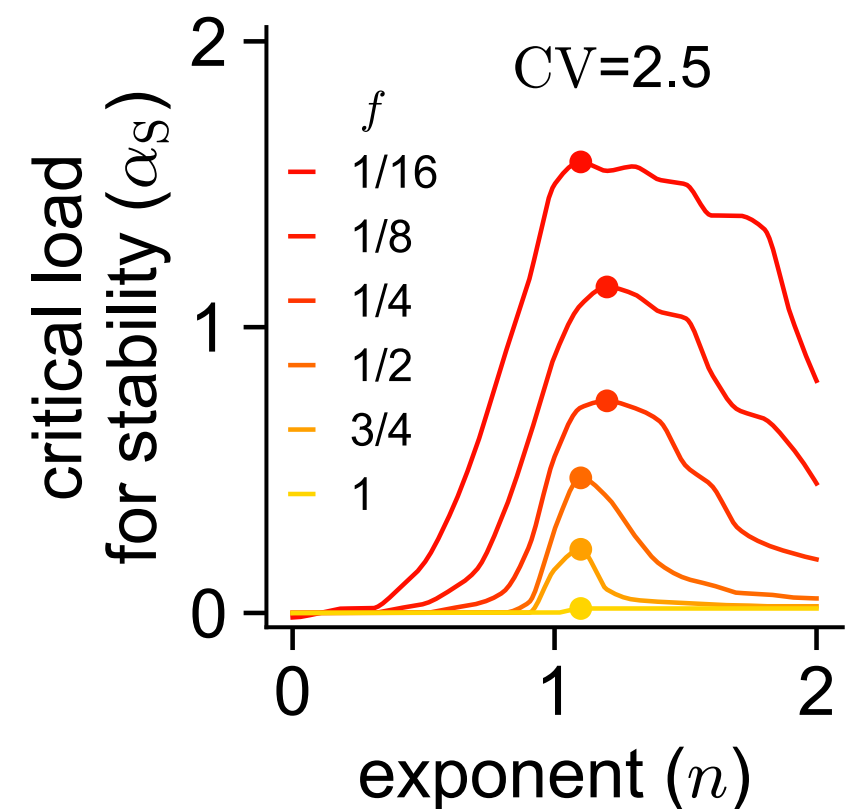
The dynamical regime of recall

max. weights sum



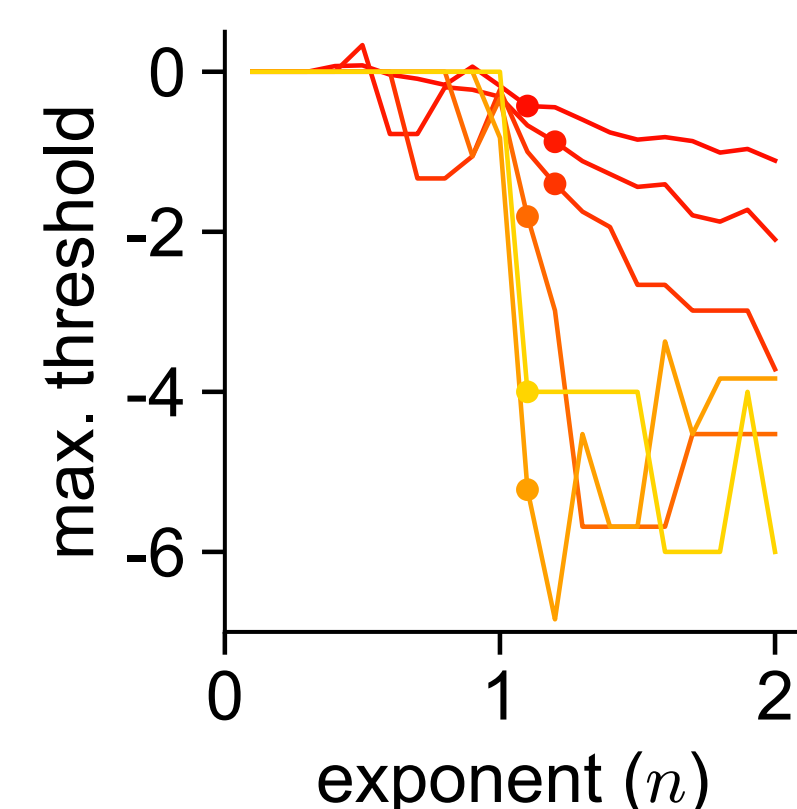
inhibition-dominated weights at $n \geq 1$

critical load for stability (α_S)



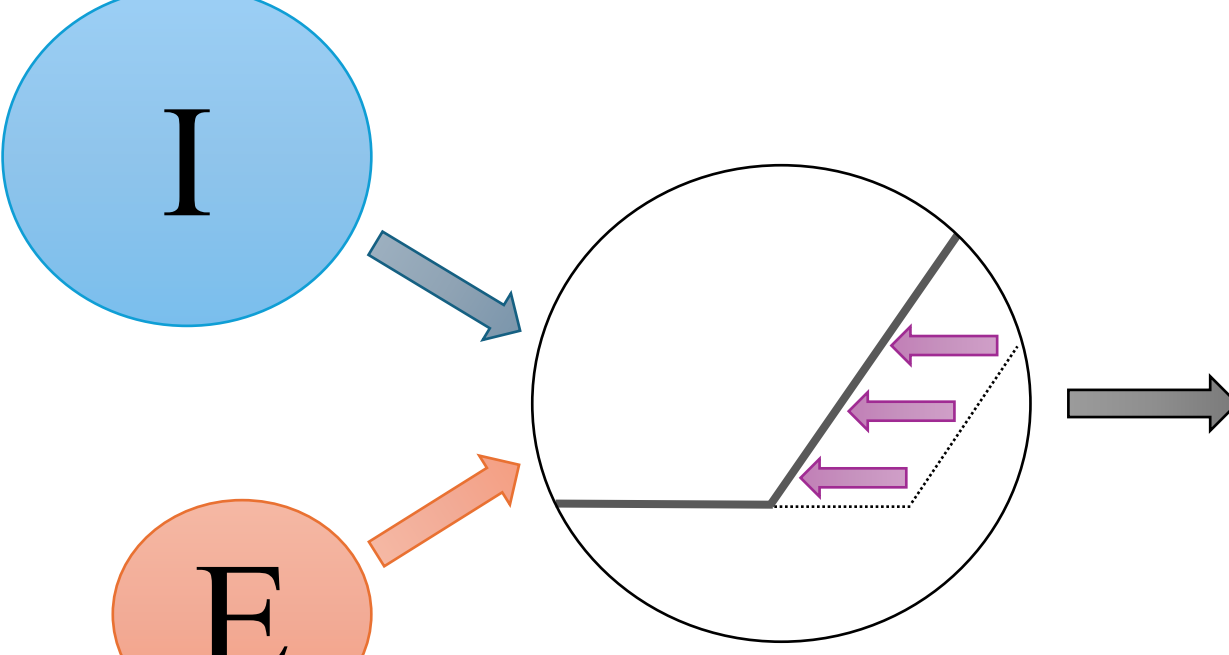
negative threshold at $n \geq 1$

max. threshold



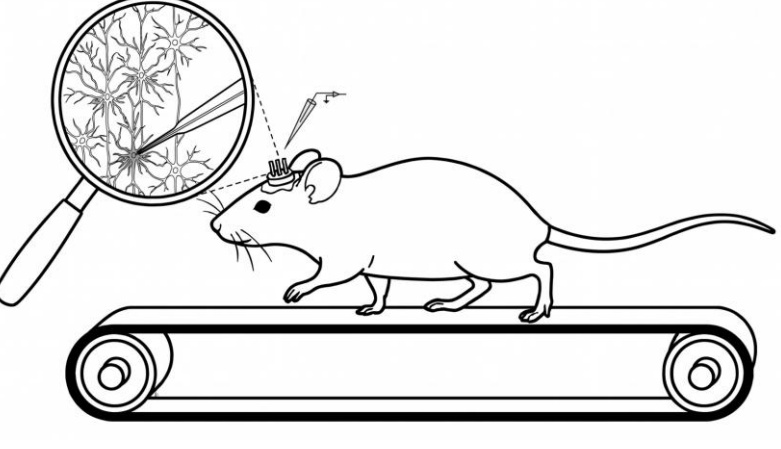
negative threshold at $n \geq 1$

predictions

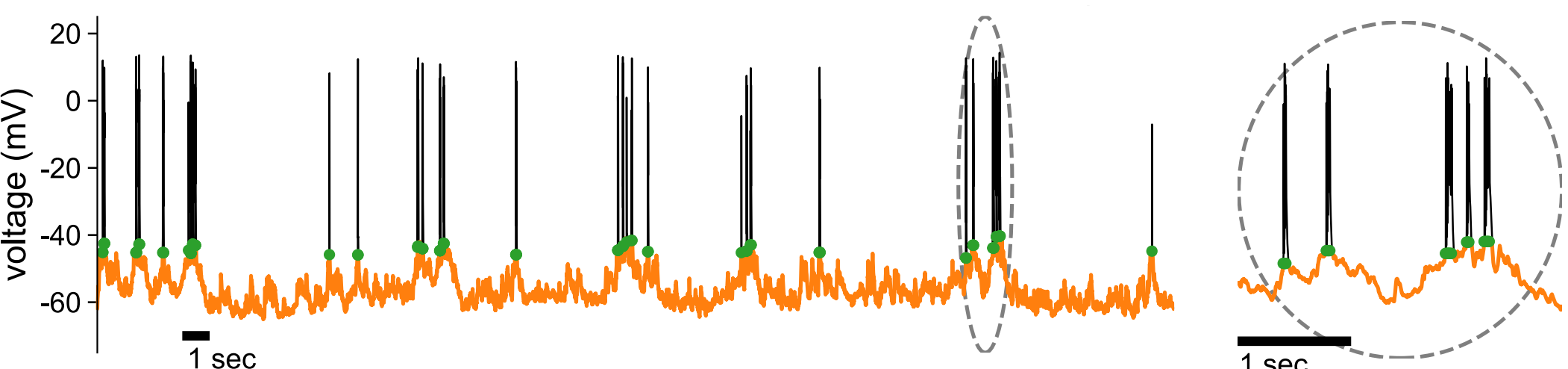


- inhibition-dominated recurrent inputs
- elevated excitability (reduced threshold)
- threshold-linear activation

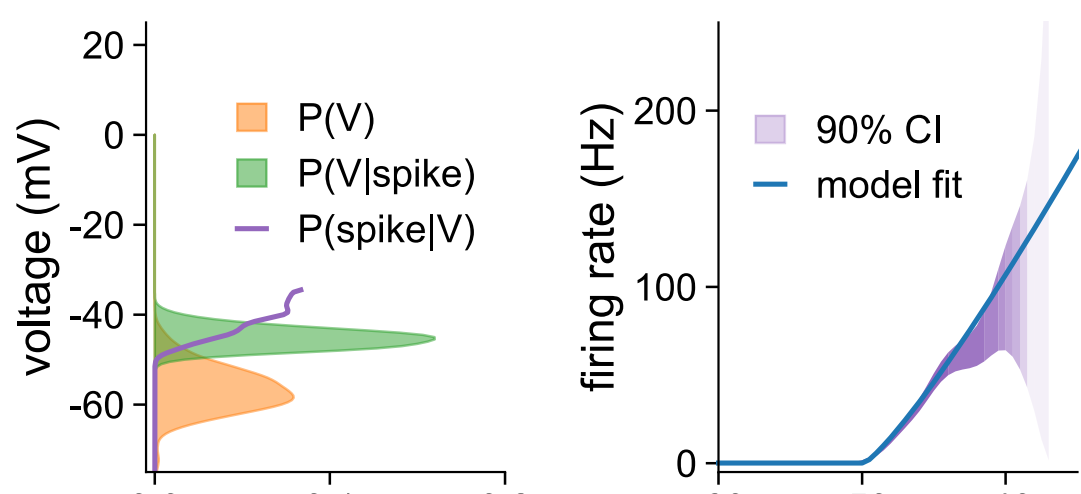
experimental results



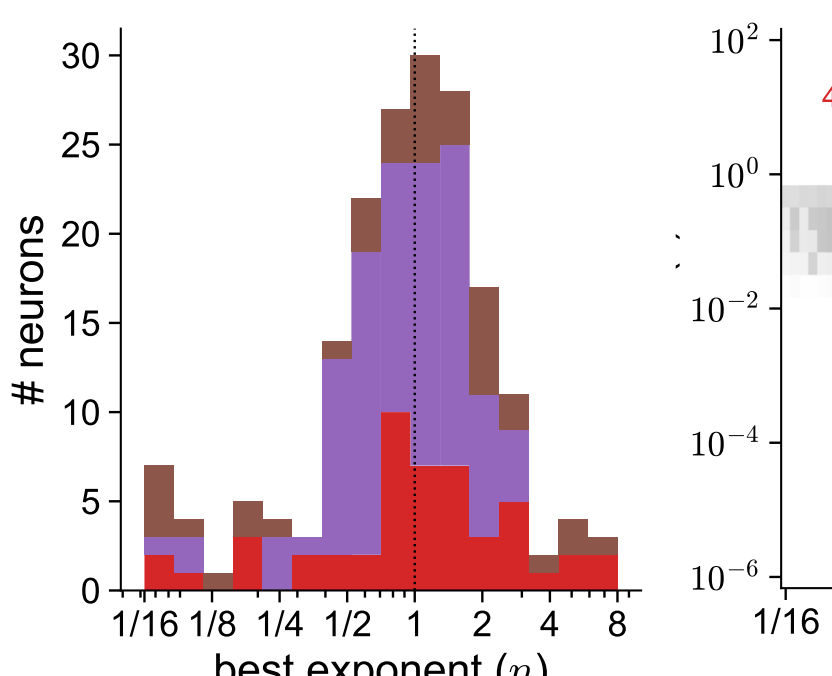
whole-cell patch-clamp recordings *in vivo*, in awake mice running on a linear belt

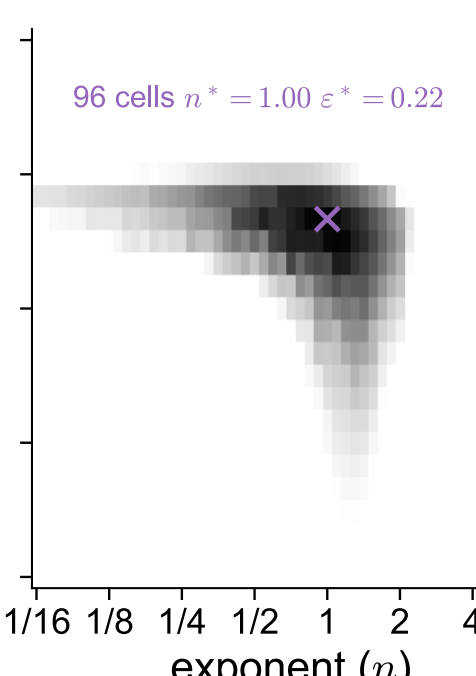
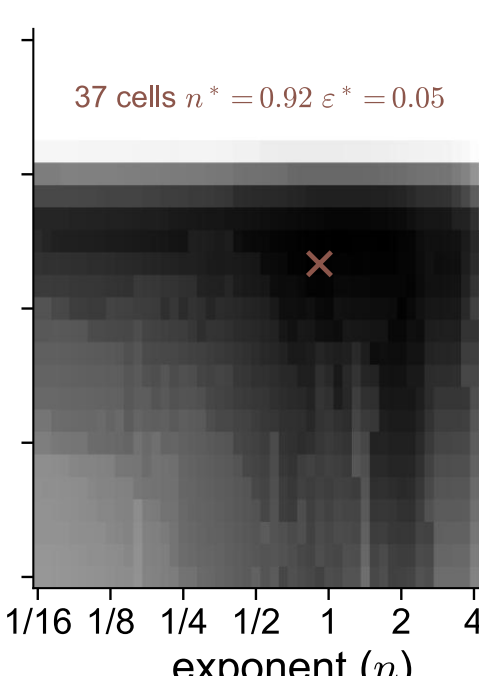
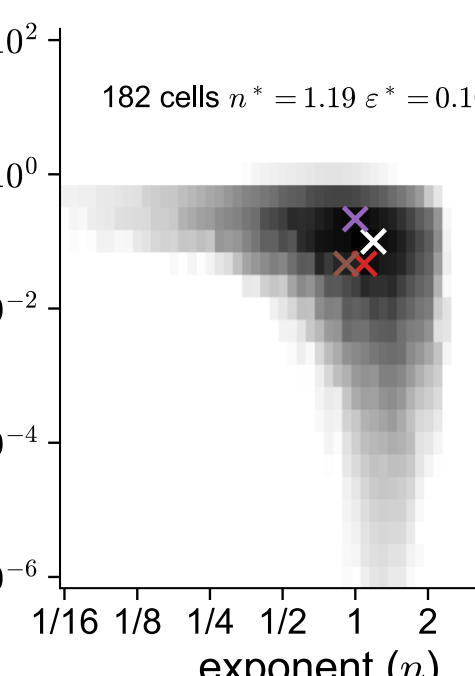


non-parametric estimation of voltage-dependent spiking probability $P(\text{spike}|V)$ and credible interval



parametric model



predictions status

- A net inhibitory response was observed in CA3 pyramidal neurons when activating recurrent collaterals [7], [8], per our “inhibition-dominated recurrent connectivity” prediction.
- CA3 pyramidal neurons fire spontaneously, without recurrent inputs [9], [10], per our “positive excitability” prediction.
- Analysis of new data from the Jonas lab and two previous studies [5], [6] finds that an exponent of ~1.2 best fits the data, per our “threshold-linear activation” prediction.

references

[1] Cohen, U., Picher, M.M., Navas-Olivé, A., & Lengyel, M. (2026). The activation function of CA3 pyramidal neurons is optimal for the stable recall of memory patterns. bioRxiv.

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[3] Mizuseki, K., & Buzsáki, G. (2013). Preconfigured, skewed distribution of firing rates in the hippocampus and entorhinal cortex. Cell Reports.

[4] Alme, C. B., Miao, C., Jezek, K., Treves, A., Moser, E. I., & Moser, M. B. (2014). Place cells in the hippocampus: eleven maps for eleven rooms. PNAS.

[5] Malezieux, M., Kees, A. L., & Mulle, C. (2020). Theta oscillations coincide with sustained hyperpolarization in CA3 pyramidal cells, underlying decreased firing. Cell Reports.

[6] Li, Y., Briguglio, J. J., Romani, S., & Magee, J. C. (2024). Mechanisms of memory-supporting neuronal dynamics in hippocampal area CA3. Cell.

[7] Spencer, W. A., & Kandel, E. R. (1961). Hippocampal neuron responses to selective activation of recurrent collaterals of hippocampofugal axons. Experimental Neurology.

[8] Sun, Q., Sotayo, A., Cazzulino, A. S., Snyder, A. M., Denny, C. A., & Siegelbaum, S. A. (2017). Proximodistal heterogeneity of hippocampal CA3 pyramidal neuron intrinsic properties, connectivity, and reactivation during memory recall. Neuron.

[9] Cohen, I., & Miles, R. (2000). Contributions of intrinsic and synaptic activities to the generation of neuronal discharges in *in vitro* hippocampus. The Journal of Physiology.

[10] Kowalski, J., Gan, J., Jonas, P., & Pernía-Andrade, A. J. (2016). Intrinsic membrane properties determine hippocampal differential firing pattern *in vivo* in anesthetized rats. Hippocampus.

